Palo Alto NGFW & Multi-Router Enterprise Lab

Comprehensive Technical Reference: Routing, Interfaces, Sessions, Policies, Backup & Lifecycle Operations

Engineer / Author:	Nishu Saini	**Lab Date:**	01-Oct-2026
Operational Session:	11:45 AM – 01:30 PM IST	**Target Platform:**	Palo Alto PAN-OS 10.x & VyOS 1.5
Firewall Instance:	PA-VM-1 (PA-VM / QEMU)	**Security Architecture:**	4-Zone Star Topology (Inside, DMZ1, Outside, DMZ2)

1. Routing Architecture: Route Table (RIB), Forwarding Information Base (FIB) & CLI Paging

Palo Alto distinguishes between the **Routing Information Base (RIB)** managed by the control plane routing engine, and the **Forwarding Information Base (FIB)** installed into the data plane hardware. Use these commands to audit route state:

```
# 1. View all active routing table entries in the control plane (RIB):
admin@PA-1> show routing route

# 2. View routes filtered by a specific Virtual Router or Type:
admin@PA-1> show routing route virtual-router default
admin@PA-1> show routing route type connect
admin@PA-1> show routing route type static

# 3. View the active Data Plane Forwarding Information Base (FIB):
admin@PA-1> show routing fib

# 4. Disable pagination pausing (prevents '--more--' or 'lines 1-23' screen hangs):
admin@PA-1> set cli pager off

# 5. Route Summary (shows total route counts and memory footprint per protocol):
admin@PA-1> show routing summary

# 6. Hardware Forwarding Lookup Simulation (tests exact egress interface and next-hop):
admin@PA-1> test routing fib-lookup ip 192.168.2.1 virtual-router default
```

Flag / Column	Technical Meaning	Operational Impact
u	Up / Active	Route is alive, installed, and actively forwarding packets in the data plane.
h	Host Route (/32)	Represents the firewall's own local interface IP address.
g	Gateway Route	Indicates a next-hop gateway address is required to forward traffic.
e	ECMP Route	Equal-Cost Multi-Path active across multiple peer uplinks.
`nexthop: 0.0.0.0`	Directly Connected	Target subnet resides directly on the physical interface without intermediate hops.

2. Enterprise Configuration Discipline: Pre-Commit Diff Auditing & Job Tracking

In production enterprise environments, changes must be audited and syntax-checked before committing to prevent accidental outages:

```
admin@PA-1> configure

# 1. Audit pending changes against the running configuration (shows + additions and - deletions):
admin@PA-1# show config diff

# 2. Perform a full syntax and dependency validation check prior to commit:
admin@PA-1# validate full

# 3. Commit changes with a descriptive audit message for compliance logging:
admin@PA-1# commit description "Configured 4 Layer-3 interfaces, zones, and ping management profile"

# 4. Track background commit, software download, or report jobs:
admin@PA-1> show jobs all
admin@PA-1> show jobs id 14
```

3. Core System Identity, Hostname, DNS, NTP & Static Management IP Setup

To establish Out-of-Band (OOB) management on a freshly deployed or factory-reset firewall, configure identity, DNS nameservers, NTP, and static IP addressing using the strict PAN-OS multi-line hierarchy:

```
admin@PA-VM> configure

# 1. System Hostname & Domain Name
set deviceconfig system hostname PA-NGFW-Lab
set deviceconfig system domain innspark.local

# 2. Timezone & System Clock Synchronization
set deviceconfig system timezone Asia/Kolkata
set deviceconfig system ntp-servers primary-ntp-server ntp-server-address time.google.com

# 3. Domain Name System (DNS) Resolvers
set deviceconfig system dns-setting servers primary 8.8.8.8 secondary 8.8.4.4

# 4. Out-of-Band (OOB) Static Management Interface Setup (Verified Working Commands)
admin@PA-VM# set deviceconfig system type static
admin@PA-VM# set deviceconfig system ip-address 172.17.0.50
admin@PA-VM# set deviceconfig system netmask 255.255.0.0
admin@PA-VM# set deviceconfig system default-gateway 172.17.0.1
admin@PA-VM# set deviceconfig system dns-setting servers primary 8.8.8.8
admin@PA-VM# commit

# 5. Management Plane Security Hardening (Disable Unencrypted Protocols)
set deviceconfig system service disable-telnet yes
set deviceconfig system service disable-http yes

exit
```

Key Takeaway / Operational Best Practice: Critical PAN-OS Syntax Rule: *In PAN-OS CLI, set deviceconfig system type static is a standalone command. Appending ip-address directly after static on the same line produces an **'Invalid syntax'** error. Always execute set deviceconfig system type static, followed by separate statements for ip-address, netmask, and default-gateway, then commit.*

4. Enterprise Data Plane: Layer 3 Interfaces, Virtual Router & Security Zones

```
admin@PA-1> configure
# ================================================================================
# 1. CREATE INTERFACE MANAGEMENT PROFILE (ALLOW PING FOR DIAGNOSTICS)
# ================================================================================
set network profiles interface-management-profile Allow-Ping ping yes

# ================================================================================
# 2. CONFIGURE LAYER 3 DATA INTERFACES & ATTACH PROFILE
# ================================================================================
# Ethernet1/1: Peer to Router-1 (INSIDE / Trust LAN)
set network interface ethernet ethernet1/1 layer3 ip 192.168.1.2/24
set network interface ethernet ethernet1/1 layer3 interface-management-profile Allow-Ping
set network interface ethernet ethernet1/1 comment "Uplink-Router1-INSIDE"

# Ethernet1/2: Peer to Router-2 (DMZ1 Segment)
set network interface ethernet ethernet1/2 layer3 ip 192.168.2.2/24
set network interface ethernet ethernet1/2 layer3 interface-management-profile Allow-Ping
set network interface ethernet ethernet1/2 comment "Uplink-Router2-DMZ1"

# Ethernet1/3: Peer to Router-3 (OUTSIDE / External)
set network interface ethernet ethernet1/3 layer3 ip 192.168.3.2/24
set network interface ethernet ethernet1/3 layer3 interface-management-profile Allow-Ping
set network interface ethernet ethernet1/3 comment "Uplink-Router3-OUTSIDE"

# Ethernet1/4: Peer to Router-4 (DMZ2 Segment)
set network interface ethernet ethernet1/4 layer3 ip 192.168.4.2/24
set network interface ethernet ethernet1/4 layer3 interface-management-profile Allow-Ping
set network interface ethernet ethernet1/4 comment "Uplink-Router4-DMZ2"

# ================================================================================
# 3. BIND INTERFACES TO VIRTUAL ROUTER 'default'
# ================================================================================
set network virtual-router default interface [ ethernet1/1 ethernet1/2 ethernet1/3 ethernet1/4 ]

# ================================================================================
# 4. CREATE SECURITY ZONES & BIND INTERFACES
# ================================================================================
set zone INSIDE network layer3 ethernet1/1
set zone DMZ1 network layer3 ethernet1/2
set zone OUTSIDE network layer3 ethernet1/3
set zone DMZ2 network layer3 ethernet1/4

commit
exit
```

5. Layer 2 ARP Cache Verification & End-to-End ICMP Diagnostics

```
# 1. Inspect the Global ARP Table:
admin@PA-1> show arp all

interface         ip              mac               status   ttl
-------------------------------------------------------------------------
ethernet1/1       192.168.1.1     0c:e1:c3:5e:00:00  c        1142
ethernet1/2       192.168.2.1     0c:bc:d4:39:00:00  c        1185
ethernet1/3       192.168.3.1     0c:ab:92:1b:00:00  c        1190
ethernet1/4       192.168.4.1     0c:86:af:4e:00:00  c        1195

# 2. Execute Bidirectional Ping Tests from Palo Alto CLI (Syntax: count BEFORE host):
admin@PA-1> ping source 192.168.1.2 count 4 host 192.168.1.1  # INSIDE (Router 1) -> 0% loss
admin@PA-1> ping source 192.168.2.2 count 4 host 192.168.2.1  # DMZ1   (Router 2) -> 0% loss
admin@PA-1> ping source 192.168.3.2 count 4 host 192.168.3.1  # OUTSIDE(Router 3) -> 0% loss
admin@PA-1> ping source 192.168.4.2 count 4 host 192.168.4.1  # DMZ2   (Router 4) -> 0% loss

# 3. Complementary Ping Verification from VyOS Routers (e.g., Router 1):
vyos@vyos# run ping 192.168.1.2 count 4
4 packets transmitted, 4 received, 0% packet loss, time 3005ms (rtt ~3.1 ms)
```

ARP Status	Technical Condition	Troubleshooting Action
`c (Complete)`	Successful Layer 2 resolution	Hardware link, carrier, and IP addressing are 100% operational.
`i (Incomplete)`	Broadcast sent; zero reply received	Check for disconnected GNS3 virtual cables, carrier down, or IP subnet mismatch.
`s (Static)`	Manually configured entry	Permanent entry programmed into hardware; does not expire.

6. Stateful Flow Inspection: Session Table Auditing & Policy Simulation

Palo Alto is a stateful NGFW. Beyond pinging, engineers must inspect active sessions and verify policy compliance:

```
# 1. View all active stateful sessions passing through the firewall:
admin@PA-1> show session all

# 2. Filter live sessions by specific source IP, destination IP, or protocol:
admin@PA-1> show session all filter source 192.168.1.1
admin@PA-1> show session all filter destination 192.168.2.1

# 3. Inspect a specific session ID in deep detail (shows NAT, bytes, packets, and decrypt state):
admin@PA-1> show session id 45892

# 4. Simulate a Security Policy match without generating live packet traffic:
admin@PA-1> test security-policy-match from INSIDE to DMZ1 source 192.168.1.1 destination 192.168.2.1 protocol 1
# (Outputs the exact rule name matched, action: allow/deny, and applied security profiles)
```

7. Virtualization Engineering: Resolving QEMU Carrier Loss & Stale uBridge Tunnels

```
# 1. Diagnosing Link State on VyOS Linux Guest:
vyos@vyos:~$ ip link show eth0
2: eth0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 ... state DOWN

# 2. Forcing Virtual Link Up via QEMU Monitor (e.g., port 42799):
(qemu) set_link virtio-net-pci.0 on

# 3. Purging Stale GNS3 uBridge Tunnel Endpoints via Control Socket (port 37833):
bridge delete QEMU-8eb8f3fd-374f-4fda-8e2f-3f2872764f78-1

# 4. Re-establishing Virtual Link via GNS3 REST API:
curl -X POST http://localhost:3080/v2/projects/{project_id}/links \
  -H "Content-Type: application/json" \
  -d '{"nodes": [{"node_id": "PA_ID", "adapter_number": 1, "port_number": 0},
                 {"node_id": "ROUTER_ID", "adapter_number": 0, "port_number": 0}]}'
```

8. Initial Configuration: 10 Core Procedures (CLI Commands & Web GUI Step-by-Step Guide)

A complete operational curriculum detailing both the precise PAN-OS CLI syntax and the exact Web GUI navigation paths for staging, accessing, securing, backing up, restoring, and managing the firewall lifecycle.

Topic 1: Initial Configuration: Basics — Lab Setup

```
# 1. Physical / Logical Topology Allocations:
# - PA ethernet1/1: 192.168.1.2/24 (Zone: INSIDE)  <---> Router 1 eth0 (192.168.1.1/24)
# - PA ethernet1/2: 192.168.2.2/24 (Zone: DMZ1)    <---> Router 2 eth0 (192.168.2.1/24)
# - PA ethernet1/3: 192.168.3.2/24 (Zone: OUTSIDE) <---> Router 3 eth0 (192.168.3.1/24)
# - PA ethernet1/4: 192.168.4.2/24 (Zone: DMZ2)    <---> Router 4 eth0 (192.168.4.1/24)
# - PA management:  172.17.0.50/16 (OOB Mgmt)     <---> NAT1 Node (docker0 172.17.0.1/16)

# 2. Initial Firewall Identity & Lab Baseline CLI Commands:
admin@PA-VM> configure
admin@PA-VM# set deviceconfig system hostname PA-VM-LAB
admin@PA-VM# set deviceconfig system domain innspark.local
admin@PA-VM# set deviceconfig system timezone Asia/Kolkata
admin@PA-VM# commit
```

Web GUI Navigation Steps:

GNS3 Canvas & Web GUI Setup Steps:

1. ***Topology Placement:*** *Drag PA-VM-1, 4 VyOS Enterprise Routers, and 1 NAT node onto the GNS3 canvas.*
2. ***Cabling:*** *Interconnect PA-VM ethernet1/0 to Router-1 eth0, ethernet1/1 to Router-2 eth0, ethernet1/2 to Router-3 eth0, ethernet1/3 to Router-4 eth0, and management to NAT1 node.*
3. ***Power On:*** *Start all nodes and verify green status LEDs.*
4. ***Web GUI Baseline Setup:*** *Navigate to* ***Device*** *→* ***Setup*** *→* ***Management*** *→ Click* ***Gear (■■)*** *on General Settings → Set Hostname: PA-VM-LAB, Domain: innspark.local → Click* ***OK*** *→* ***Commit****.*

Topic 2: Initial Configuration: Basics — How to Access Palo Alto Firewall

```
# 1. Console Access (Serial / GNS3 Telnet Port 5008):
$ telnet 127.0.0.1 5008
PA-VM login: admin
Password: admin (or Nishu@123@)

# 2. Secure Shell (SSH) Access over Management IP:
$ ssh admin@172.17.0.50 (or ssh admin@192.168.1.1)

# 3. Mandatory First-Login Password Change Prompt:
Enter old password: admin
Enter new password: <SecurePassword123>
Confirm password:   <SecurePassword123>
```

Web GUI Navigation Steps:
Web GUI Access Steps:
1. Connect PC Ethernet adapter to the MGT port and configure static IP 172.17.0.10/16 (or 192.168.1.10/24).
*2. Open Chrome or Firefox and navigate to **https://172.17.0.50** (or https://192.168.1.1).*
*3. Bypass SSL certificate warning by clicking **Advanced** → **Proceed to 172.17.0.50 (unsafe)**.*
4. Enter administrator credentials to log into the Palo Alto Dashboard.

Topic 3: Initial Configuration: Basics — How to Enable Telnet

```
# A. Enable Telnet on Management (MGT) Port:
admin@PA-1> configure
admin@PA-1# delete deviceconfig system service disable-telnet
admin@PA-1# commit

# B. Enable Telnet on Data Ports (via Interface Management Profile):
admin@PA-1# set network profiles interface-management-profile Allow-Telnet telnet yes
admin@PA-1# set network interface ethernet ethernet1/1 layer3 interface-management-profile Allow-Telnet
admin@PA-1# commit
```

Web GUI Navigation Steps:
Web GUI Steps:
***On MGT Port:** Go to **Device** → **Setup** → **Management** → Click **Gear (■■)** on Management Interface Settings → Check **Telnet** → Click **OK** → **Commit**.*
***On Data Ports:** Go to **Network** → **Network Profiles** → **Interface Mgmt** → Click **Add** → Name: Allow-Telnet → Check **Telnet** → Click **OK**. Then go to **Network** → **Interfaces** → Click target port → **Advanced** → Select Management Profile: Allow-Telnet → **Commit**.*

Topic 4: Initial Configuration: Basics — How to Enable SSH

```
# A. Enable SSH on Management (MGT) Port (Enabled by default):
admin@PA-1> configure
admin@PA-1# delete deviceconfig system service disable-ssh
admin@PA-1# commit

# B. Enable SSH on Data Ports (via Interface Management Profile):
admin@PA-1# set network profiles interface-management-profile Allow-SSH ssh yes
admin@PA-1# set network interface ethernet ethernet1/1 layer3 interface-management-profile Allow-SSH
admin@PA-1# commit
```

Web GUI Navigation Steps:
Web GUI Steps:
***On MGT Port:** Go to **Device** → **Setup** → **Management** → Click **Gear (■■)** on Management Interface Settings → Verify **SSH** checkbox is enabled → Click **OK** → **Commit**.*
***On Data Ports:** Go to **Network** → **Network Profiles** → **Interface Mgmt** → Click **Add** → Check **SSH** → Bind to interface under **Network** → **Interfaces** → **Commit**.*

Topic 5: Initial Configuration: Basics — How to Enable HTTP

```
# A. Enable Plaintext HTTP on Management (MGT) Port (Disabled by default for security):
admin@PA-1> configure
admin@PA-1# delete deviceconfig system service disable-http
admin@PA-1# commit

# B. Enable HTTP on Data Ports (via Interface Management Profile):
admin@PA-1# set network profiles interface-management-profile Allow-HTTP http yes
admin@PA-1# set network interface ethernet ethernet1/1 layer3 interface-management-profile Allow-HTTP
admin@PA-1# commit
```

Web GUI Navigation Steps:
Web GUI Steps:
On MGT Port: *Go to* ***Device*** *→* ***Setup*** *→* ***Management*** *→ Click* ***Gear (■■)*** *on Management Interface Settings → Check* ***HTTP*** *checkbox → Click* ***OK*** *→* ***Commit****.*
On Data Ports: *Go to* ***Network*** *→* ***Network Profiles*** *→* ***Interface Mgmt*** *→ Click* ***Add*** *→ Check* ***HTTP*** *→ Attach profile to interface →* ***Commit****.*

Topic 6: Initial Configuration: Basics — How to Enable HTTPS

```
# A. Enable Secure HTTPS on Management (MGT) Port (Enabled by default):
admin@PA-1> configure
admin@PA-1# delete deviceconfig system service disable-https
admin@PA-1# commit

# B. Enable HTTPS Web GUI on Data Ports:
admin@PA-1# set network profiles interface-management-profile Allow-HTTPS https yes
admin@PA-1# set network interface ethernet ethernet1/1 layer3 interface-management-profile Allow-HTTPS
admin@PA-1# commit
```

Web GUI Navigation Steps:
Web GUI Steps:
On MGT Port: *Go to* ***Device*** *→* ***Setup*** *→* ***Management*** *→ Click* ***Gear (■■)*** *on Management Interface Settings → Ensure* ***HTTPS*** *is checked → Click* ***OK*** *→* ***Commit****.*
On Data Ports: *Go to* ***Network*** *→* ***Network Profiles*** *→* ***Interface Mgmt*** *→ Check* ***HTTPS*** *→ Apply profile to interface →* ***Commit****.*

Topic 7: Initial Configuration: Basics — How to Verify Connectivity

```
# 1. Physical & Logical Interface Verification:
admin@PA-1> show interface all
admin@PA-1> show interface ethernet1/1

# 2. Layer 2 ARP Resolution Table:
admin@PA-1> show arp all

# 3. Layer 3 Routing Table & FIB Audit:
admin@PA-1> show routing route
admin@PA-1> show routing fib

# 4. Bidirectional Ping Tests:
admin@PA-1> ping source 192.168.1.2 count 4 host 192.168.1.1

# 5. Route Simulation Lookup:
admin@PA-1> test routing fib-lookup ip 192.168.2.1 virtual-router default
```

Web GUI Navigation Steps:
Web GUI Steps:
1. ***Interface State:*** *Go to* ***Network*** *→* ***Interfaces*** *→ Verify green link LEDs, IP addresses, and Zones.*
2. ***ARP Table:*** *Go to* ***Network*** *→* ***Virtual Routers*** *→ Click default → Click* ***More Runtime Stats*** *→ Select* ***ARP*** *tab.*
3. ***Routing Table:*** *Go to* ***Network*** *→* ***Virtual Routers*** *→ Click default → Click* ***More Runtime Stats*** *→ Select* ***Routing Table*** *tab.*
4. ***Traffic Monitoring:*** *Go to* ***Monitor*** *→* ***Logs*** *→* ***Traffic*** *to observe live sessions and flow state.*

Topic 8: Initial Configuration: Basics — How to Take Backup

```
# Method A: Save Named Snapshot Locally on Flash:
admin@PA-1> configure
admin@PA-1# save config to Backup_Lab_20261001.xml
admin@PA-1# exit

# Method B: Export Configuration to Remote SCP Server:
admin@PA-1> scp export configuration from running-config.xml to user@192.168.1.100:/backups/

# Method C: Export Configuration to Remote TFTP Server:
admin@PA-1> tftp export configuration from running-config.xml to 192.168.1.100
```

Web GUI Navigation Steps:
Web GUI Steps:
*1. Navigate to **Device** → **Setup** → **Operations**.*
*2. Under Configuration Management, click **Save named configuration snapshot** → Enter name Backup_20261001.xml → Click **OK**.*
*3. Under Export, click **Export named configuration snapshot** → Select running-config.xml (or your named snapshot) → Click **OK** → Download `.xml` file to your PC.*

Topic 9: Initial Configuration: Basics — How to Reset With Factory Default

```
# Method A: Soft Reset via Running CLI:
admin@PA-1> request system private-data-reset
# System prompts: 'Do you want to continue? (y or n)'. Press 'y'.
# Firewall formats /dev/vda6, purges all policies/IPs, and reboots to default 192.168.1.1.

# Method B: Automated Boot to Maintenance Mode:
admin@PA-1> debug system maintenance-mode on
admin@PA-1> request restart system

# Method C: Hard Reset via Console Maintenance Mode:
# Connect serial console (9600-8-N-1) -> Reboot -> Type 'maint' at autoboot prompt -> Select 'Factory Default' -> Reboot.
```

Web GUI Navigation Steps:
Web GUI Steps:
*1. Navigate to **Device** → **Setup** → **Operations**.*
*2. Under Device Operations, click **Revert to default factory settings**.*
*3. Click **Yes** to confirm the warning prompt.*
4. The firewall will format user partitions, wipe all certificates/rules, and reboot to factory IP 192.168.1.1.

Topic 10: Initial Configuration: Basics — How to Restore Config

```
# Method A: Load from Locally Saved Snapshot (CLI):
admin@PA-1> configure
admin@PA-1# load config from Backup_Lab_20261001.xml
admin@PA-1# commit
admin@PA-1# exit

# Method B: Merge Partial Configuration:
admin@PA-1# load config partial mode merge from Backup_Lab_20261001.xml from-xpath ...
```

Web GUI Navigation Steps:
Web GUI Steps:
*1. Navigate to **Device** → **Setup** → **Operations**.*
*2. Under Configuration Management, click **Import named configuration snapshot** → Click **Browse** → Select your backup `.xml` file → Click **OK**.*
*3. Click **Load named configuration snapshot** → Select the imported `.xml` file from dropdown → Click **OK**.*
*4. Click **Commit** at the top-right to apply and activate the restored configuration.*

9. Maintenance Mode Architecture & Factory Reset Internals (The 'tune2fs' Phase)

```
# 1. Inspecting Remote Maintenance Boot Flag:
admin@PA-1> debug system maintenance-mode show
Maintenance-mode boot is disabled

# 2. Instructing Firewall to Boot Directly into Maintenance Mode on Next Reboot:
admin@PA-1> debug system maintenance-mode on
Maintenance-mode boot is enabled for next reboot

# 3. Triggering System Reboot:
admin@PA-1> request restart system

# 4. Checking Management Server Daemon Status (Confirms Readiness for Login):
admin@PA-1> show management-server status
# (Outputs: running - confirms mgmtsrvr is ready, eliminating 'Login incorrect')

# 5. Monitoring Disk Space Across Partitions:
admin@PA-1> show system disk-space
```

Key Takeaway / Operational Best Practice: Why PA-VM Appears Frozen on 'tune2fs -c /dev/vda6' During Factory Reset:
*Partition /dev/vda6 stores all firewall logs, session state databases, and reporting structures. During a factory reset, the kernel formats and rebuilds this multi-gigabyte volume from scratch. In virtualized environments (QEMU), this takes between **5 to 10 minutes** with zero console output. **Never power off or reset the VM during this stage**, as interrupting it corrupts the virtual disk filesystem. Once completed, the firewall boots normally and presents the PA-VM login: prompt.*